

SMRP Scholarship Application

PROGRAM DEADLINE : July 08 2023 at 11:59 PM EDT (Midnight)

Applicant Information

Name: Jacob Vaught
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App ID: 04239214
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General Information

Contact Information

First Name	Jacob
Last Name	Vaught
Street Address	211 Main St. Apt 111
City	Columbiass
State/Province	SC
Postal/ Zip Code	29210
Country	United States
Phone Number	803-645-9962
Email	jvaught@sc.edu
How did you hear about our scholarship program?	Other
I am a	Junior
Are you enrolled in a four- year university or a graduate program?	Yes

Please note you are eligible for all four scholarships as a junior in a four year program. Please select which scholarships you would like to apply for below. You may select more than one.

SMRPCO Dorothy and Jack Nicholas
Scholarship ~SMRP Legacy of Excellence
Scholarship~SMRP Scholarship

Academic Information

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Name of Institution	University of South Carolina
Major	Mechanical Engineering & Electrical Engineering
Street Address	300 Main St
City	Columbia
State/Province	SC
Postal/ Zip Code	29210
Country	United States
Contact Name at Institution	
Contact Title	
Contact Phone Number	
Contact Email Address	
GPA	3.59

Leadership

Leadership

List clubs, student sections of professional societies, and any leadership positions that you are involved in.

Leadership:

ASME (American Society of Mechanical Engineers) - Vice President

IEEE (Institute of Electrical and Electronics Engineers) - President

SAE (Society of Automotive Engineers) - President

Pi Tau Sigma - Secretary

SAMPE (Society for the Advancement of Material and Process Engineering) - Secretary

Rocketry Club - Co-Founder, Secretary ('22-'23), Senior Advisor ('23-Present)

115th Student Senate - Senator (Finance Committee, Academics Committee, Student Life Committee)

Membership:

SHPE (Society of Hispanic Professional Engineers)

NSBE (National Society of Black Engineers)

SWE (Society of Women Engineers)

ACM (Association for Computing Machinery)

AIAA (American Institute of Aeronautics and Astronautics)

SME (Society of Manufacturing Engineers)

ASCE (American Society of Civil Engineers)

CEC Peer Mentor

Include a copy of your transcript (official or unofficial) of courses completed to date. Please do not include your name on this if possible. This is to ensure anonymity.

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Write a summary of work experience in the maintenance and reliability field (500 word max).

Throughout my career, I have gained varied experience in the engineering field, with several roles closely aligning with the maintenance and reliability field.

As a Laboratory Manager under Dr. Austin Downey, I have built my expertise in developing Finite Element Analysis (FEA) models for vibration analysis of complex structures in structural health monitoring. This experience is directly applicable to the maintenance and reliability field as such models are crucial for predicting system behavior and identifying potential failures. In this role, I also managed a group of over 30 individuals, overseeing lab maintenance and supply orders.

In my role as a Summer Intern at Westinghouse Electric Company LLC, I took on a project to design and model a component to prevent lateral movement of a torque testing rig. This work directly contributed to improving the reliability of the rig. Additionally, I developed Python programs for analyzing grid growth in three nuclear fuel assemblies, focusing on Vogtle Unit 2. This analysis could contribute to understanding and enhancing the reliability of nuclear power generation. Furthermore, I was involved in a preliminary analysis of the potential for switching from 18-month to 24-month fuel cycles, showing my capacity to consider strategic, long-term efficiency and reliability in operations.

While working on a project to increase the enrichment value of nuclear fuel, I utilized CAD (SOLIDWORKS) to calculate POSITY (porosity) of a Combustion Engineering™ grid strap. This experience allowed me to assess and improve product design for better reliability.

These roles have not only enabled me to refine my technical skills but also to grow my leadership and management abilities. I believe that these experiences make me well-suited for a role in the maintenance and reliability field.

**Include resume (one page max),
please do not list your name on this
if possible to ensure anonymity.** Jacob Vaught Resume-Current.pdf

Write a statement (500 words) explaining why you should be awarded the scholarship.

As an engineer who has dedicated his academic and professional journey to pushing the boundaries of what is possible, I am excited about the opportunity to apply for this scholarship. I am confident that my passion for innovation, my academic accomplishments, and my hands-on experience in various aspects of engineering make me a deserving candidate.

I've been drawn to the intricacies of engineering from a young age, fueling my pursuit of degrees in Electrical Engineering, Computer Science, and Mechanical Engineering from the University of South Carolina. Throughout my academic career, I have maintained a strong GPA, earned accolades such as making the Dean's and President's Lists multiple times, and graduated Magna Cum Laude in my 2-year technical degree. These accomplishments reflect my dedication and commitment to excel in my chosen field.

But for me, it's not just about academic excellence; it's about applying what I've learned to make tangible contributions to the field. As a Laboratory Manager and Undergraduate researcher under Dr. Austin Downey, I have demonstrated my ability to apply complex concepts to real-world situations, developing Finite Element Analysis models for vibration analysis. This work aligns closely with maintenance and reliability, as these models are integral to predicting system behavior and identifying potential failures.

I carried this ethos to my role as a Summer Intern at Westinghouse Electric Company, where I designed a component to improve the reliability of a torque testing rig, and developed Python programs for nuclear fuel assembly analysis. I also contributed to a project that could significantly reduce nuclear fuel costs for consumers, demonstrating my commitment to not only innovative engineering solutions, but also to creating value for the wider community.

In addition to my technical contributions, I believe in fostering strong leadership and collaborative skills. As President of IEEE and SAE, and Secretary of ASME and SAMPE, I have worked to build more engaging, inclusive, and productive academic communities. These experiences have taught me how to motivate teams, plan effectively, and make strategic decisions – skills that are vital for any successful engineer.

Receiving this scholarship would allow me to continue my research without stressing over financial aspects and make further contributions to the engineering field. I have been awarded multiple grants for my research in Laser Powder Bed Fusion Additive Manufacturing under Dr. Lang Yuan. This scholarship would provide the necessary funding to deepen my research, enabling me to potentially revolutionize how we approach additive manufacturing.

I believe that I am a deserving candidate for this scholarship due to my academic accomplishments, my dedication to practical applications of engineering principles, and my commitment to leadership and community. I am not just passionate about engineering; I am devoted to pushing the field forward, making the world a better place through innovation. Given the opportunity, I am eager to leverage this scholarship to continue my research and contribute significantly to the engineering community.
